From Plant-Level Traditional Use to Compound-Level Target Activity: Leakage-Controlled Multiple-Instance Learning for Vietnamese Medicinal Plants

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Abstract

Background: Traditional-medicine databases connect plants to reported uses and plants to constituent metabolites, whereas modern bioactivity databases measure activity at the compound–target level. Naively assigning every plant-level use to every reported metabolite creates false labels and obscures which chemical evidence, if any, supports the use. We study whether Vietnamese ethnopharmacological annotations can improve molecular-target ranking without making this cross-granularity assumption.

Methods: We link an authorized, versioned VietHerb snapshot to measured activities from ChEMBL 37 for human DPP4, acetylcholinesterase, and COX-2. Each medicinal plant is represented as a bag of standardized metabolite instances. A shared molecular encoder is trained with masked compound-level activity objectives and an attention multiple-instance objective for curated plant-level therapeutic concepts. Missing traditional-use annotations are handled as unlabelled rather than confirmed negative. Evaluation uses globally deduplicated chemical out-of-distribution and botanical cold-start splits, five training seeds, early recognition metrics, and dependence-aware paired bootstrap intervals. Controls include activity-only multi-task learning, naive label broadcasting, shuffled bag labels, random auxiliary labels, and natural-product importance weighting.

Results: [RESULT PENDING: Insert frozen cohort counts, primary paired PR-AUC difference, confidence interval, per-target effects, early-recognition metrics, and falsification-control outcomes.]

Conclusions: [RESULT PENDING: State a conclusion only after the locked test analysis. A null result must be reported as such.]

1 Introduction

Natural products remain an important source of chemical diversity for drug discovery, but their biological evidence is heterogeneous and frequently recorded at a different level from the molecule being screened. VietHerb was created to organize Vietnamese medicinal plants together with metabolites, therapeutic effects, morphology, and geography. Its original release reported 2,881 species, 10,887 metabolites, and many-to-many species–metabolite and species–effect relations [Nguyen-Vo et al., 2019]. This local resource is scientifically valuable, but a reported use of a plant is neither evidence that every constituent is active nor proof of clinical efficacy.

Prior computational ethnopharmacology already links traditional symptoms, efficacies, herbs, ingredients, and molecular targets. HTINet and HTINet2 use heterogeneous networks and knowledge graphs for herb–target prediction [Wang et al., 2019, Duan et al., 2024]; other systems use clinical symptom profiles or multimodal ingredient–target information [Cheung, 2026, Liang et al., 2025]. Consequently, the broad claim that traditional knowledge has not been used for target prediction is untenable. The unresolved issue addressed here is narrower: VietHerb annotations are plant-level labels, while ChEMBL observations are compound-level measurements. This is

naturally a multiple-instance learning (MIL) problem, in which a plant is a bag and its reported metabolites are instances [Ilse et al., 2018, Doran and Ray, 2012].

Reliable evaluation is equally important. Ligand benchmarks can reward memorization when close analogues cross partitions [Wallach and Heifets, 2018], and conventional scaffold splitting does not eliminate every form of chemical redundancy. Traditional-medicine data add botanical leakage because the same metabolite can occur in several species. Missing compound–target and plant–use edges also cannot automatically be interpreted as negatives, making positive–unlabelled (PU) learning a relevant sensitivity analysis [Kiryo et al., 2017, Wang et al., 2026].

We therefore make four contributions. First, we define an auditable data model that preserves compound, plant, therapeutic-concept, assay, and provenance relations. Second, we formulate cross-granularity supervision with attention MIL instead of label broadcasting. Third, we introduce split manifests that keep identical structures, related chemistry, and plant components disjoint. Fourth, we isolate the value of ethnopharmacological information with matched architectures and falsification controls. The preregistered primary question is whether plant-bag supervision improves compound-level out-of-distribution PR-AUC beyond activity-only multi-task learning.

2 Materials and methods

2.1 Data sources and governance

VietHerb provides plant–metabolite and plant–therapeutic-effect relations [Nguyen-Vo et al., 2019]. Because no explicit database redistribution licence was found during our audit, the confirmatory analysis requires a maintainer-authorized export. Raw rows will not be redistributed without written permission. Published artifacts will otherwise contain code, schemas, source identifiers, checksums, aggregate counts, and split-generation logic. Traditional-use records are treated as hypothesis-generating cultural and historical evidence, not as validated biomedical claims.

Measured activities are extracted from the versioned ChEMBL 37 SQLite release; the live REST service is used only for exploratory engineering. ChEMBL is a manually curated, large-scale bioactivity resource with explicit assay and document provenance [Zdrazil et al., 2024]. LOTUS may be used only for an optional occurrence/provenance sensitivity analysis [Rutz et al., 2022]. Aggregated natural-product databases are not assumed to provide independent activity ground truth [Chandrasekhar et al., 2025].

2.2 Target-activity cohort

The pilot targets are human dipeptidyl peptidase 4 (DPP4, CHEMBL284), acetylcholinesterase (AChE, CHEMBL220), and prostaglandin G/H synthase 2 (COX-2/PTGS2, CHEMBL230). The primary cohort contains high-confidence, single-protein biochemical binding assays with standardized IC_{50} or K_i values in nM. Functional assays form a separate sensitivity cohort. Records with data-validity warnings, duplicate flags, unresolved structures, or ambiguous target assignments are excluded. Exact standardized structure–target replicates are aggregated by median pChEMBL; pairs spanning at least one log unit are flagged as contradictory and excluded from the primary analysis.

The prespecified binary endpoint is active at pChEMBL ≥ 6. Thresholds 5 and 7 and continuous pActivity models are sensitivity analyses. Unmeasured compound–target pairs remain masked. All stereochemistry and structure standardization decisions, rejected rows, release metadata, and SHA-256 hashes are retained in a machine-readable manifest.

2.3 Therapeutic-use normalization

Original text is preserved. Candidate mappings are retrieved from frozen MeSH/Mondo releases, after which a language model may rank candidates but may not accept a label automatically. The stored record includes source span, ontology version, candidate list, confidence, model/prompt hash,

abstention, and reviewer decisions. Two pharmacognosy or pharmacy reviewers independently annotate a prespecified stratified sample before adjudication. We report exact and ancestor-aware precision, recall, F1, raw agreement, and Cohen's κ; sample size is selected for a target confidence-interval width rather than fixed solely by convenience.

The first model uses three coarse, clinically interpretable families: metabolic/diabetes, cognition/neurological use, and pain/inflammation. ATC is not used as the primary disease ontology because it classifies active drug substances rather than diseases or symptoms.

2.4 Molecular and botanical split construction

Molecules are standardized globally and assigned once across all targets and plants. Two locked regimes answer different questions. The chemical-OOD regime groups molecules by a prespecified fingerprint clustering rule and reports train–test nearest-neighbour Tanimoto distributions. A scaffold component split is retained as a reproducible secondary regime. The botanical cold-start regime holds out plant taxa and explicitly handles metabolites shared between taxa.

For the strict joint split, molecule nodes are joined when they share a chemical group or occur in the same plant. Entire transitive connected components are assigned to train, validation, or test. Thus, if compound A is chemically linked to B and B occurs with C in one species, all three remain in one partition. No feature transformation, loss weight, class prior, or hyperparameter is fitted using test data.

2.5 Multiple-instance multi-task model

Let x_{ij} denote metabolite j reported in plant i, and let $h_{ij} = f_\theta(x_{ij})$ be its shared molecular representation. The primary heads predict measured compound–target activities with masked, class-weighted binary cross entropy. For each plant bag, gated attention computes

$$e_{ij} = w^\top \left[\tanh(Vh_{ij}) \odot \sigma(Uh_{ij})\right], \tag{1}$$

$$a_{ij} = \frac{\exp(e_{ij})}{\sum_k \exp(e_{ik})}, \tag{2}$$

$$z_i = \sum_j a_{ij} h_{ij}, \tag{3}$$

followed by one plant-use head per therapeutic family [Ilse et al., 2018]. Attention weights provide instance attribution but are not interpreted as causal explanations.

An observed plant use is positive and absence is unlabelled. Auxiliary heads therefore use non-negative PU risk [Kiryo et al., 2017]. For class prior π, positive risk R_p^+, positive-as-negative risk R_p^-, and unlabelled negative risk R_u^-, the unbiased estimate is

$$R_{\mathrm{PU}} = \pi R_p^+ + R_u^- - \pi R_p^-, \tag{4}$$

with the non-negative correction applied when the empirical negative component falls below zero. Priors are frozen from training-only estimation and varied in sensitivity analyses. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{activity}} + \lambda \mathcal{L}_{\text{plant-MIL}}. \tag{5}$$

The compute-feasible confirmatory encoder is a radius-2, 2,048-bit chiral Morgan fingerprint followed by a two-layer MLP. This deliberately separates the scientific contribution from an expensive foundation model. A D-MPNN/Chemprop baseline [Yang et al., 2019] and one pretrained molecular encoder are secondary comparisons using identical split IDs.

2.6 Baselines and falsification controls

Required models are: per-target fingerprint logistic regression; random forest or gradient boosting; separate neural target models; shared activity-only multi-task learning; naive plant-label broadcasting; the proposed MIL model; and optional domain importance weighting. Controls shuffle use labels among training plants while preserving prevalence, replace uses with equally sized random auxiliary labels, compare expert-verified with confidence-weighted mappings, and equalize parameter counts and update counts. The primary comparison is MIL versus the architecture-matched activity-only model. Naive broadcasting is an ablation, not the proposed method.

2.7 Training and statistical analysis

The MLP uses hidden dimensions 1,024 and 512, dropout 0.25, AdamW with learning rate 3×10^{-4} and weight decay 10^{-4}, batch size 256, and early stopping patience 12 on validation macro PR-AUC. Auxiliary weights $\{0.1, 0.3, 1.0\}$ are selected on validation data. Seeds are 13, 37, 71, 101, and 137.

The primary endpoint is the paired change in macro test PR-AUC. We additionally report target-wise PR-AUC and prevalence, BEDROC ($\alpha = 20$), EF1% and EF5%, precision/recall at fixed experimental budgets, AUROC, Brier score, and calibration. PR-AUC is emphasized for imbalanced screening [Saito and Rehmsmeier, 2015], while BEDROC measures early recognition [Truchon and Bayly, 2007]. Paired 95% intervals use 1,000 bootstrap replicates with structure cluster as the resampling unit for compound endpoints and plant as the unit for bag endpoints. Target-level secondary tests use Benjamini–Hochberg correction.

3 Results

3.1 Cohort and shift audit

[RESULT PENDING: Insert the full data-flow diagram and counts after each exclusion.]

Table 1: Frozen cohort summary. Values remain blank until the confirmatory ChEMBL 37 and authorized VietHerb build passes every acceptance gate.

Endpoint	Measurements	Molecules	Active (%)	Test positives
DPP4	–	–	–	–
AChE	–	–	–	–
COX-2	–	–	–	–
Plant-use bags	–	–	–	–

Report structural overlap, nearest-neighbour similarity, bag sizes, metabolites per plant, plants per metabolite, label prevalence, and the number/size of connected components. Reconcile the live VietHerb counts with the historical database paper rather than assuming that they are identical.

3.2 Primary predictive comparison

[RESULT PENDING: Insert all five seeds and the paired, cluster-bootstrap confidence interval for MIL minus activity-only macro PR-AUC.]

3.3 Ablations, cold starts, and interpretation

[RESULT PENDING: Report per-target negative transfer, botanical cold-start results, PU-prior sensitivity, auxiliary-weight sensitivity, and learning curves.] Attention examples may be used to generate hypotheses only after checking their stability across seeds and chemically similar instances.

Table 2: Locked chemical-OOD test performance. Report mean across seeds with a paired 95% confidence interval; never report the best seed alone.

Model	Macro PR-AUC	BEDROC20	EF1%	Brier
Single-task fingerprint	–	–	–	–
Activity-only MTL	–	–	–	–
Naive broadcast	–	–	–	–
VietHerb-MIL	–	–	–	–
Shuffled-bag control	–	–	–	–

4 Discussion

The principal result must be interpreted as incremental predictive value under the locked evaluation, not validation of traditional medicine. If MIL improves over activity-only learning but not over shuffled labels, the gain is consistent with generic regularization rather than semantic ethnopharmacological signal. If naive broadcasting performs similarly to MIL, chemical or botanical leakage and label granularity require further investigation. If all ethnomedical variants underperform, negative transfer is an informative result and should not be hidden by selecting favorable targets.

Docking or short molecular dynamics may provide a later structure-based plausibility assessment, but neither constitutes biological validation. Docking scores have well-known limitations for affinity ranking [Warren et al., 2006]. Prospective biochemical assays with positive, negative, interference, and purity controls are required to claim target inhibition.

Limitations include incomplete or changing plant databases, uncertain traditional-use normalization, incomplete chemical inventories for each plant, assay heterogeneity, class-prior uncertainty in PU learning, and the possibility that attention is not a faithful explanation. Written data permission and a domain collaborator are release gates rather than administrative afterthoughts.

5 Conclusion

[RESULT PENDING: Write after the confirmatory analysis. The allowed conclusion is bounded to molecular prioritization under the evaluated shifts.]

Data and code availability

Code, configurations, schemas, tests, source locks, and derived manifests are available at `https://github.com/MinhNhat-2504/vietherb-mtl`. ChEMBL-derived artifacts follow CC BY-SA 3.0. VietHerb row-level data will be released only if the maintainers provide explicit permission; otherwise the repository will publish acquisition instructions, hashes, identifiers, and reproducible transformation code without redistributing restricted rows.

Author contributions, funding, and competing interests

[RESULT PENDING: Complete before submission, including the pharmacy/pharmacognosy reviewers and any experimental collaborators.]

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